

KHI / Kawasaki Heavy Industries — LH2 Solution Assessment Database

Prepared for: Gentari (PETRONAS group) — LH2 value chain workstream **Derived from:** Gentari ⇌ KHI technical questionnaire (Feb–Apr 2026) and KHI technical/commercial decks (Apr–May 2026), plus KHI corporate materials. **Scope:** Gate-to-gate KHI-offered technology across liquefaction, loading terminal, seaborne transport, and receiving/regasification. Production (upstream H2) and end-use/reconversion are out of KHI's offer and out of scope here. **Units & currency:** SI/metric; H2 mass in kg; costs in the source's native unit with cost-year caveats stated per **CLAUDE.md §5**. **Status:** This is a structured **database derived from KHI's assessment responses**, not a reproduction of the questionnaire itself. Every figure below is traceable to the source register in §10. Values KHI declined to disclose are recorded as explicit gaps (§9), never estimated.

1. How to read this document

Marker	Meaning
(cited)	Backed by a specific KHI or IAE document, listed in §10
[ASSUMPTION]	A value we chose deliberately to make the database usable; rationale given
[GAP — not disclosed]	KHI was asked and declined, pending a Feasibility Study

All figures are KHI's own representations of their technology and are **not independently verified** by Gentari. Where KHI gave a range instead of a point value, the range is carried, not collapsed.

2. Executive summary

- KHI positions **LH2 as commercially and technically viable for earlier introduction than commonly assumed**, relative to NH3 and MCH carriers — this is KHI's stated view, not an independent finding (*cited, KHI2604 R1*).
- The **core technology is proven at pilot scale** (Suiso Frontier, 1,250 m³, 2021–2022 Australia–Japan voyage) and is **scaling through a 40,000 m³ vessel under construction** toward a **160,000 m³ commercial-scale target** (*cited, kawasaki-hydrogen-activities, kawasaki-2026-questionnaire*).
- Liquefaction SEC is 8–9 kWh/kg-H2** (*cited*), with material efficiency gains expected only in the **2035–2040** window, not near-term.
- KHI reports **no hydrogen loss** across its plant concept and **no filling restrictions or sloshing-driven BOG penalty** on the carrier — both are KHI's stated design targets rather than operating-data-backed guarantees (*cited, reply Q15/Q20*).
- The single largest gap for investment screening is cost:** KHI has **not disclosed CapEx or OpEx** in USD/kg for liquefaction, terminals, or the carrier newbuild — in each case KHI's answer was "a Feasibility Study is necessary" (§9). The only cost data available is a **second-hand, unlabelled-axis IAE cost stack** reproduced in a KHI slide (§7), which is unit- and cost-year-unverified.

- **Regulatory/permitting detail for LH2 import terminals outside Japan is not yet available from KHI**; only a qualitative LH2-vs-NH3 hazard comparison was given (§8).

3. Liquefaction

Boundary: gaseous H₂ in → LH₂ at ≈20 K, ≈1 bar, out.

Parameter	Value	Basis	Source
Specific energy consumption (SEC)	8–9 kWh/kg-H₂	varies with liquefier scale & boundary conditions	(cited, kawasaki-2026-questionnaire)
Plant-level power consumption	0.55 kWh/Nm³	KHI fleet/plant assumption used in cost-basis slides	(cited, kawasaki-2026-supplemental)
Unit train capacity	115 t/day per train	scales toward ~150 tpd in KHI's "Tech" (large-scale/high-speed) case	(cited, kawasaki-2026-supplemental)
Train count — Base scale	7 trains	Base = 2.5 billion Nm ³ /y	(cited, kawasaki-2026-supplemental)
Train count — Large scale	27 trains	Large = 10 billion Nm ³ /y	(cited, kawasaki-2026-supplemental)
Minimum economical scale	≈10,000 tpa	depends on target-market offtake pricing	(cited, kawasaki-2026-questionnaire)
Liquefaction cycle	H₂ Claude cycle + N₂ precooling	proven track record	(cited, kawasaki-2026-questionnaire)
Precooling alternative	Hydrocarbon mixed refrigerant	under consideration; not yet proven	(cited, kawasaki-2026-questionnaire)
Ortho–para (O–P) conversion	Performed inside the liquefier , catalytic, to planned para-H ₂ fraction	—	(cited, reply Q6)
O–P catalyst	Fe-based (in service); Co-based under evaluation	no expected degradation → no replacement planned	(cited, reply Q7)
Liquefier performance degradation	None expected over operating life	—	(cited, reply Q16)
Downstream BOG handling (pre-loading)	Re-liquefied — via ejector in the liquefier, or compressor with cryo-rated suction	—	(cited, reply Q3/Q14)

Parameter	Value	Basis	Source
Hydrogen loss, chain-wide	0% — KHI's plant concept assumes no H2 losses	KHI design claim, not measured field data	(cited, reply Q15)
SEC improvement outlook	Significant improvement expected 2035–2040	driven by subcomponents currently under development, not near-term	(cited, reply Q1)
CapEx, USD/kg	[GAP — not disclosed]	KHI: "Feasibility Study necessary... varies with plant size"	reply Q8
OpEx, USD/kg	[GAP — not disclosed]	same caveat	reply Q9
Carbon intensity, kgCO ₂ e/kg-H2 (with RE supply)	[GAP — not disclosed]	same caveat	reply Q11
PFD / heat & mass balance	[GAP — not disclosed]	KHI: shareable "at Feasibility Study timing"	reply Q10
Permitting (Japan & abroad)	Approved case-by-case via a special evaluation committee in Japan for matters outside existing law; similar country-specific approach expected elsewhere	qualitative only	(cited, reply Q12)

4. Loading / export terminal

Boundary: LH2 from liquefier → LH2 loaded onto carrier.

Parameter	Value	Basis	Source
Boil-off rate (BOR) at rest	0.1 %/day	at-rest storage	(cited, kawasaki-2026-supplemental)
Storage tank size	64,000 m³/tank	export side	(cited, kawasaki-2026-supplemental)
Tank count — Base / Large	3 tanks / 4 tanks	Base 2.5B Nm³/y, Large 10B Nm³/y	(cited, kawasaki-2026-supplemental)
Dead volume	Broadly similar to LNG tanks — driven by pump NPSHr, not by in-tank pump presence	—	(cited, reply Q4)
Dead-volume minimization option	Spherical tank + pressurised send-out (no pumps)	a small "keep-cool" heel may remain	(cited, reply Q4)
Loading arm — demonstrated size	6-inch, TRL 6–7	pilot demonstration completed on Suiso Frontier	(cited, reply Q19)
Loading arm — commercial size	16-inch, not yet verified	verification expected in ~3 years from Apr 2026; both arm types by TBG	(cited, reply Q19 supplemental)

Parameter	Value	Basis	Source
Export-terminal CapEx, USD/kg	[GAP — not disclosed]	only the aggregate IAE cost stack (\$7) is available	—

5. Shipping / carrier

Boundary: LH2 loaded → LH2 discharged at import terminal.

Parameter	Value	Basis	Source
Demonstration vessel (Suiso Frontier)	1,250 m³ , 116 m LOA, 13 knots, diesel-electric, 25 crew	world's first LH2 carrier; Dec 2021–Feb 2022 Australia → Japan voyage	(cited, kawasaki-hydrogen-activities)
Vessel under construction	40,000 m³	same cargo-tank technology as Suiso Frontier; no major issues flagged	(cited, reply Q29)
Commercial-scale target vessel	160,000 m³ at 29.6 km/h (≈16 knots)	KHI cost-basis slide	(cited, kawasaki-2026-supplemental)
Fleet count — Base / Large	2 ships / 8 ships	Base 2.5B Nm³/y, Large 10B Nm³/y	(cited, kawasaki-2026-supplemental)
BOG management (propulsion)	BOG used as fuel via a dual-fuel (DF) engine KHI is developing	fuel-gas consumption rate = BOR	(cited, reply Q23/Q32)
Onboard reliquefaction	Not adopted — too much electrical energy and space required	—	(cited, reply Q33)
Filling-ratio / sloshing restriction	None — no filling-ratio limit, no sloshing impact on BOG	—	(cited, reply Q20)
Max sailing distance	Limited only by the MGO fuel tank , not by BOG	—	(cited, reply Q21)
100% MGO (no BOG-as-fuel) regime	Tank pressure holds within MARVS for several days ; GCU burns excess BOG if MARVS is exceeded	—	(cited, reply Q26)
Loading/unloading duration	≈1–1.5 days per vessel call	varies with terminal facilities	(cited, reply Q27)
Preferred shipyard	KHI Sakaide Shipyard	lead time longer than LNG carriers — more complex double-wall tanks/piping	(cited, reply Q28)
Scale-up challenge to 160,000 m³	Cargo-tank manufacturability flagged as the key issue	—	(cited, reply Q29)
Cost premium vs. LNG carrier	Significantly higher — double-wall tanks <i>and</i> piping increase material/welding; limited supplier base for H2 cargo equipment	qualitative only, no \$/vessel figure disclosed	(cited, reply Q30/Q31)

Parameter	Value	Basis	Source
Newbuild cost, USD/vessel (160,000 m ³)	[GAP — not disclosed]	same as above	reply Q31
Power Take-Off (PTO)	Optional , per shipowner's operational policy	—	(cited, reply Q24)
FuelEU / ECA compliance	Needs higher H2 fuel fraction + green H2 to meet FuelEU; SOx met via H2/MGO; NOx Tier 3 needs SCR	—	(cited, reply Q22)
Floating Storage Unit (FSU) / ship-to-ship	Not yet specifically considered — too few LH2 carriers in service today; KHI believes the technology is applicable in principle (similar to other liquefied-gas carriers)	—	(cited, reply Q25)
Minimum heel (temperature hold)	No general figure — varies with voyage length	—	(cited, reply Q34)

6. Receiving terminal & regasification

Boundary: LH2 in → gaseous H2 at delivery pressure, out.

Parameter	Value	Basis	Source
Boil-off rate (BOR) at rest	0.1 %/day	at-rest storage, same as export side	(cited, kawasaki-2026-supplemental)
Storage tank size	65,000 m³/tank	receiving side	(cited, kawasaki-2026-supplemental)
Tank count — Base / Large	3 tanks / 11 tanks	Base 2.5B Nm ³ /y, Large 10B Nm ³ /y	(cited, kawasaki-2026-supplemental)
Vaporiser type	Open-Rack Vaporiser (ORV) , seawater heat source	large heat source required to vaporize LH2	(cited, reply Q17)
Alternative heat source	Dedicated heat-source system	if nearby aquaculture constrains cold-seawater discharge	(cited, reply Q17)
Regas heat duty	≈3.8 MJ/kg-LH2	—	(cited, reply Q18)
FSU / ship-to-ship	Same as \$5 — not yet specifically considered	—	(cited, reply Q25)
Receiving-terminal CapEx, USD/kg	[GAP — not disclosed]	only aggregate IAE cost stack available	—

7. Chain-wide cost stack (IAE basis, via KHI)

Source & caveats (read before using): KHI reproduces a per-component carrier-cost comparison attributed to IAE (Institute of Applied Energy), Gigaton Workshop, WHTC 2019 (6 May 2019) — this is **second-hand**; the primary IAE

publication is not held in this repo, and the **axis on KHI's slide is unlabelled**. Magnitudes are consistent with the METI/IAE ¥/Nm³ hydrogen supply-cost framing used elsewhere in Japanese hydrogen policy work, so the unit is carried as **JPY/Nm³, ~2019 vintage**, but this is **[needs source: primary — unit/cost-year unverified]**. **Do not use these figures for a USD/kg LCOH without verifying the unit and cost-year against the primary IAE source.**

Three scale/technology scenarios, deal volumes per KHI:

- **Base** = 2.5 billion Nm³/y
- **Large scale** = 10 billion Nm³/y
- **Tech** = 10 billion Nm³/y **with** a technology step-up: larger liquefier trains (115 → 150 tpd), higher-speed vessel (16 → 25 knot), larger storage tank (up to 200,000 m³)

Component (JPY/Nm ³ , ~2019)	LH2 Base	LH2 Large	LH2 Tech
Production (H2 supply)	10.1	10.1	9.3
Liquefaction	10.3	10.0	8.6
Loading terminal	5.6	1.9	1.8
Seaborne transport	4.0	4.0	2.3
Receiving terminal	6.3	3.4	2.8
Cracking / dehydrogenation	0.0	0.0	0.0
Other / distribution	1.3	1.3	1.3
TOTAL	37.6	30.7	26.1

For reference, KHI's slide also carries the equivalent **MCH** and **NH3** stacks at Base/Large scale (same source and unit caveats) — reproduced here because they frame KHI's own LH2-vs-alternatives argument in §8:

Component (JPY/Nm ³ , ~2019)	MCH Base	MCH Large	NH3 Base	NH3 Large
Production/Synthesis feed	10.3	10.3	10.9	10.9
Conversion (hydrogenation/synthesis)	5.5	5.4	10.3	9.6
Loading terminal	0.4	0.2	0.4	0.2
Seaborne transport	4.0	3.9	3.2	3.2
Receiving terminal	1.0	0.7	1.3	1.0
Cracking / dehydrogenation	12.7	12.7	8.9	8.6
Other / distribution	1.3	1.3	1.3	1.3

Reading, per KHI's own framing: LH2 has **no reconversion-step cost** (cracking/dehydrogenation = 0), unlike NH3 and MCH, and its total cost **falls faster with scale** (Base 37.6 → Large 30.7 → Tech 26.1) than the other carriers shown. This is KHI's argument for LH2 competitiveness — carry it as KHI's position, not as an independently verified LCOH comparison, given the unit/cost-year caveat above.

8. LH2 vs. NH3 vs. MCH — qualitative position (KHI's view)

Dimension	LH2	NH3	MCH
Toxicity	None	Toxic — needs hazard abatement (water-spray, gas detectors, evacuation zones) at receiving terminal	Toluene toxicity at carrier stage
Reconversion step at destination	None	Cracking + purification required	Dehydrogenation + purification required
Energy/process penalty	Extreme refrigeration (~20 K) + high electrical power for liquefaction	Ammonia synthesis + cracking energy	Hydrogenation + dehydrogenation energy
Safety footprint	Non-toxic but flammable high-pressure gas → larger safety exclusion/separation zones	Toxic-gas evacuation zones	Toluene flammability/toxicity
Energy efficiency (well-to-x)	Reported 5–35% lower than comparator pathways depending on stage (see note)	—	—
CO ₂ during marine transport	Can approach zero during navigation by using BOG as carrier propulsion fuel	—	—
KHI's overall position	"LH2 is not as lacking in competitiveness nor as challenging as often assumed" — advocated for earlier introduction , not a rejection of NH3	KHI: not dismissing ammonia as a future fuel	—

Note on the energy-efficiency deltas (5–35% figures): these are attributed by KHI to *IEA Global Hydrogen Review 2022*; the CO₂ comparison is attributed to *IEA, April 2023, "Towards hydrogen definitions based on their emissions intensity."* Both are **cited by KHI, second-hand**; primary IEA documents are not yet held in this repo (*cited, kawasaki-2026-comparison*).

Ammonia-as-fuel technical constraints (KHI's supporting context, not LH2-specific): ammonia combustion needs larger combustors for stable, complete combustion; large gas turbines with >10 closely-spaced combustors cannot simply swap to ammonia-compatible combustors without redesign — cited from a Feb 2025 Hanwha/Baker Hughes ammonia gas-turbine JDA example (*cited, kawasaki-2026-comparison*).

9. Disclosure-gap register (critical for investment screening)

These are figures Gentari asked for and KHI **explicitly declined to provide** pending a Feasibility Study. They are **not zero, not estimated — they are open**:

#	Gap	Segment	KHI's stated reason
1	CapEx, USD/kg	Liquefaction	"cost varies depending on the plant size"
2	OpEx, USD/kg	Liquefaction	same
3	Carbon intensity, kgCO ₂ e/kg (with RE)	Liquefaction	same

#	Gap	Segment	KHI's stated reason
4	PF ₂ / heat & mass balance	Liquefaction	shareable only "at Feasibility Study timing"
5	Newbuild cost, USD/vessel (160,000 m ³)	Shipping	premium over LNG only qualified qualitatively
6	Export/receiving terminal CapEx, USD/kg	Terminals	not separately broken out; only the aggregate, unit-unverified IAE stack exists
7	LH ₂ import-terminal permitting detail outside Japan	Cross-cutting	"does not yet have detailed regulatory/permit information for LH ₂ import terminals"
8	Malaysia government subsidy/support specifics for NH ₃ /LH ₂	Commercial (Gentari ↔ KHI mutual Q&A)	left unanswered in both questionnaire rounds

Implication for any techno-economic model built on this database: technical KPIs (energy, losses, fleet sizing, BOR) can be modeled today directly from KHI's disclosed figures. A full **USD/kg LCOH cannot be produced from KHI data alone** — CapEx/OpEx must come from a Feasibility Study, a comparable OEM benchmark (with the user's explicit permission per **CLAUDE.md §4** source hierarchy), or be supplied as a tagged **[ASSUMPTION]** at the point of use.

10. KHI corporate / track-record snapshot

- Kawasaki Heavy Industries, Ltd. — FY ended 31 Mar 2023: revenue **¥1,725.6 billion (≈USD 11,920.7 million)**; **38,254 employees**; 103 consolidated subsidiaries (*cited, kawasaki-hydrogen-activities*).
- LH₂ track record: **Suiso Frontier** (world's first LH₂ carrier, 1,250 m³) completed a demonstration voyage Hastings, Australia → Kobe, Japan, Dec 2021–Feb 2022, under the CO₂-free Hydrogen Energy Supply-chain Technology Research Association (**HySTRA**) pilot project (*cited, kawasaki-hydrogen-activities*).
- KHI's own scale-up roadmap: pilot (1,250 m³ × ~1) → commercialization demonstration (~2025, 2,500 m³-class) → commercial chain (~2030, 160,000 m³ × 2 carriers) → further scale-up toward 2050, with **indicative CIF cost figures of ¥170/Nm³ (pilot) → ¥30/Nm³ (2030 commercial) → ¥20/Nm³ (~2050)** (*cited, kawasaki-hydrogen-activities*) — **note: this CIF figure is a different KHI-internal roadmap estimate from the IAE stack in §7; do not mix the two without reconciling basis/vintage.*
- Full supply-chain scope KHI markets: production (electrolysis/fertiliser- plant hydrogen) → liquefaction → LH₂ carrier/tank/container → loading arm → end-use (hydrogen gas turbines, gas engines, fuel-cell trains, hydrogen boilers) (*cited, kawasaki-hydrogen-activities*).

11. Source register

ID	Document	Date	Tier
kawasaki-2026-questionnaire	Gentari → KHI Liquid H ₂ Questionnaire, KHI reply (Rev.1)	31 Mar / 6 Apr 2026	Proprietary
kawasaki-2026-questionnaire-r0	Same questionnaire, earlier reply (superseded)	4 Mar 2026	Proprietary

ID	Document	Date	Tier
kawasaki-2026-supplemental	Supplemental Information — Comparison between LH2 and other Carriers of Hydrogen (KHI2605 R0)	May 2026	Proprietary
kawasaki-2026-comparison	Comparison between LH2 and other Carriers of Hydrogen (KHI2604 R1)	Apr 2026	Proprietary
kawasaki-hydrogen-activities	Kawasaki's Hydrogen Activities (corporate overview)	May 2024	Proprietary
iae-2019-gigaton	Institute of Applied Energy, Gigaton Workshop, WHTC 2019	6 May 2019	Institutional (second-hand via KHI)

Full machine-readable register with file paths: `data/references.csv` . Full per-parameter database (CSV, tidy schema): `data/properties/` , `data/vessels/` , `data/costs/` — see `data/README.md` for the schema and tag rules. This PDF is a synthesized, narrative rendering of those same tables for reading and sharing; the CSVs remain the source of truth for modeling.

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